Basic Genome Structure and Sequence Exploration Using Galaxy

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Course and Section	BIO 300 - B
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Instructors	Ma'am Lilibeth Bucol & Sir Abner Bucol
Species Name	*Cynopterus brachyotis*

Part 1 — Reconfirm the Genome Identity

Species name	*Cynopterus brachyotis*
NCBI Assembly accession	GCA_009793145.1
Assembly level (if known)	Scaffold
FASTA filename in Galaxy	Cynopterus_brachyotis_genome_original.fna.gz
Genome source/database	NCBI
Approximate file size	547.1 MB

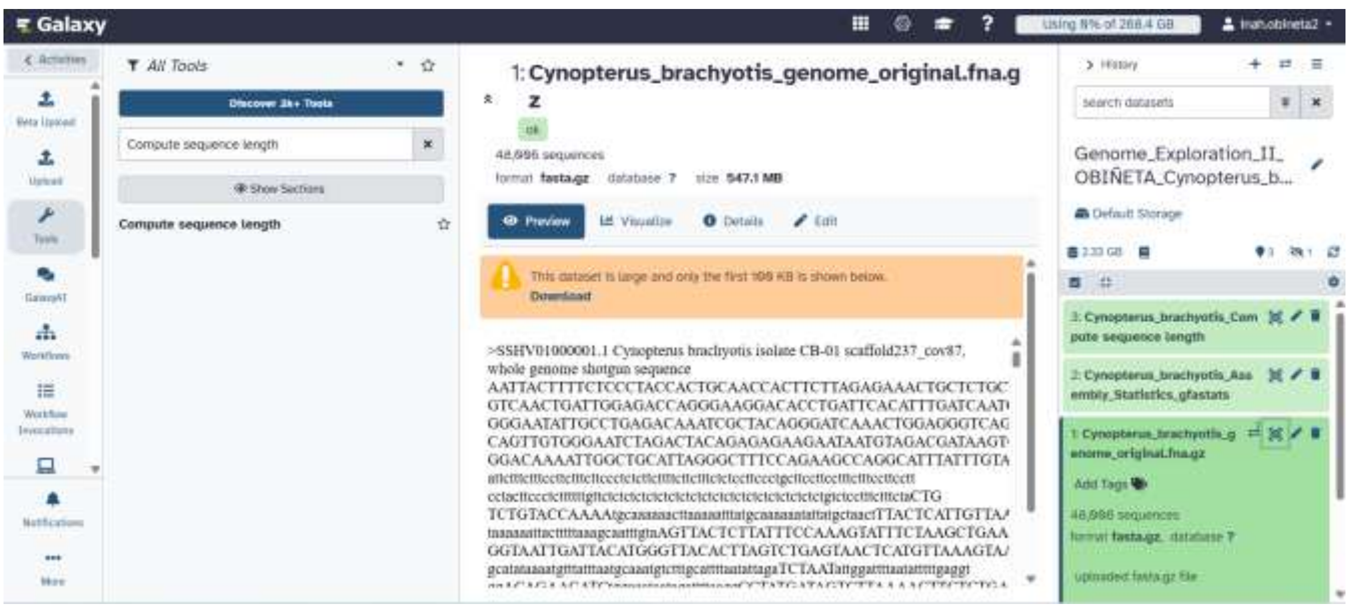

Figure 1: *Cynopterus brachyotis* Genome dataset uploaded in Galaxy History

Part 2 — Generate Basic Assembly Statistics

Metric	Values	Interpretation
Total assembly length	1,758,935,687	1.76 Gb — reasonable for a bat genome (typically 1.7–2.2 Gb).
Number of sequences/scaffolds	48,006	High count suggests a fragmented, draft-level assembly.
Minimum sequence length	1,000	Shortest scaffolds are very small, likely low-information fragments.
Maximum sequence length	4,468,567	4.5 Mb largest scaffold — far below chromosome size, confirming a draft assembly.
Mean/average sequence length	36,639.91	Much lower than the max, showing most sequences are short.
N50	251,278	Moderate contiguity — half the genome is in scaffolds ≥251 kb.
L50	1,873	Genome is spread across many scaffolds, not a few large ones.
GC content (%)	38.98	Normal for mammals (38–42%); no unusual base composition.

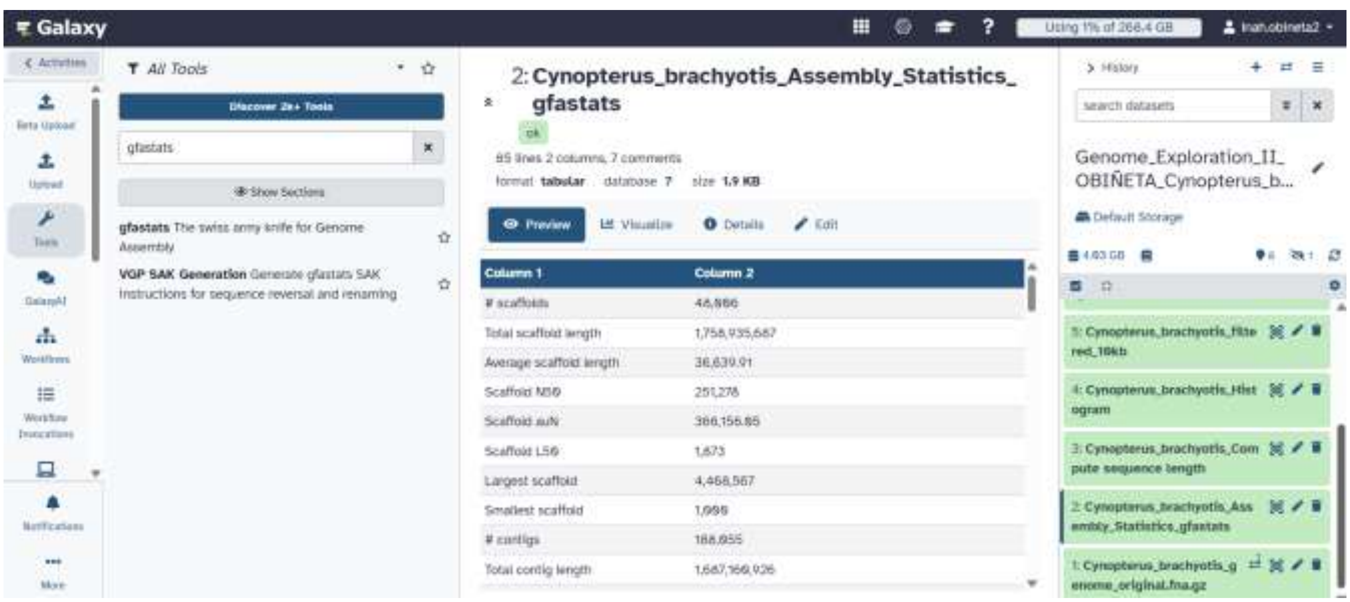

Figure 2: *Cynopterus brachyotis* Statistics output using gfasats in Galaxy History

Part 3 — Examine Sequence-Length Structure

Rank	Sequence ID	Length (bp)
1	SSHV01002152.1	4,468,567
2	SSHV01001336.1	3,130,546
3	SSHV01001042.1	2,249,068
4	SSHV01002552.1	2,005,528
5	SSHV01001936.1	1,948,660

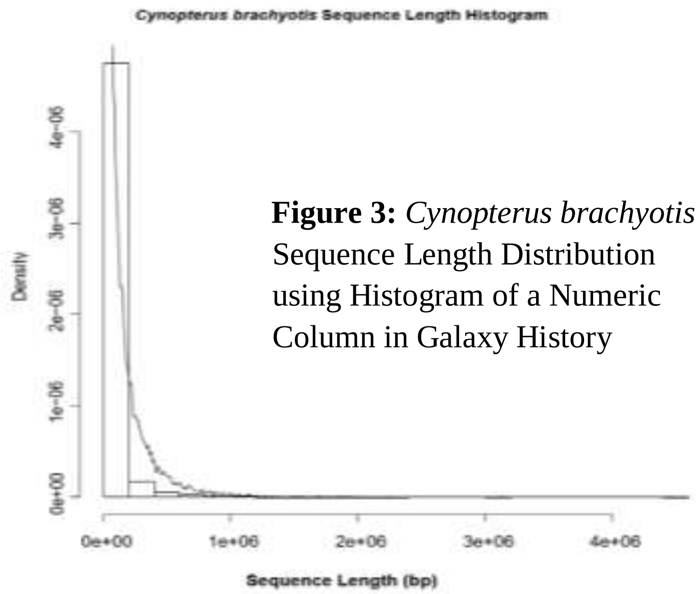

Figure 3: *Cynopterus brachyotis* Sequence Length Distribution using Histogram of a Numeric Column in Galaxy History

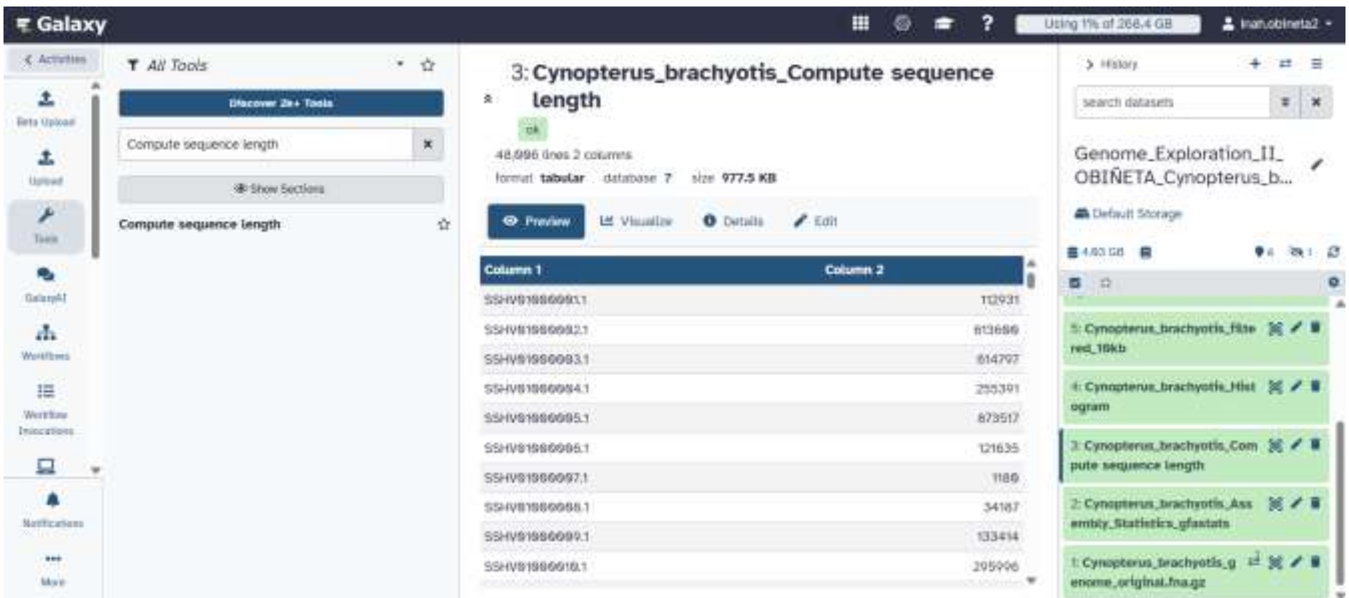

Figure 4: *Cynopterus brachyotis* Sequence Length Output using Compute Sequence Length in Galaxy History

Describe the assembly pattern. Is the genome represented by a few very long sequences, many short sequences, or a mixture?

- The genome shows a mixture, but strongly skewed toward short sequences. The top 5 scaffolds range from 1.9–4.5 million bp, but these are exceptions with 48,006 total sequences and a mean length of only 36,640 bp, most sequences are far shorter. The L50 of 1,873 confirms that half the genome's length comes from nearly 2,000 scaffolds, not a handful of large ones. This pattern is typical of a draft-quality, non-chromosome-level assembly that remains fragmented into thousands of variably-sized scaffolds.

Part 4 — Perform a Simple Length-Filtering Experiment

Metric	Original genome	After ≥10 kb filter
Total length	1,758,935,687	1,677,409,341
Number of sequences	48,006	12,483
Maximum length	4,468,567	4,468,567
N50	251,278	267,263
GC content	38.98	38.60

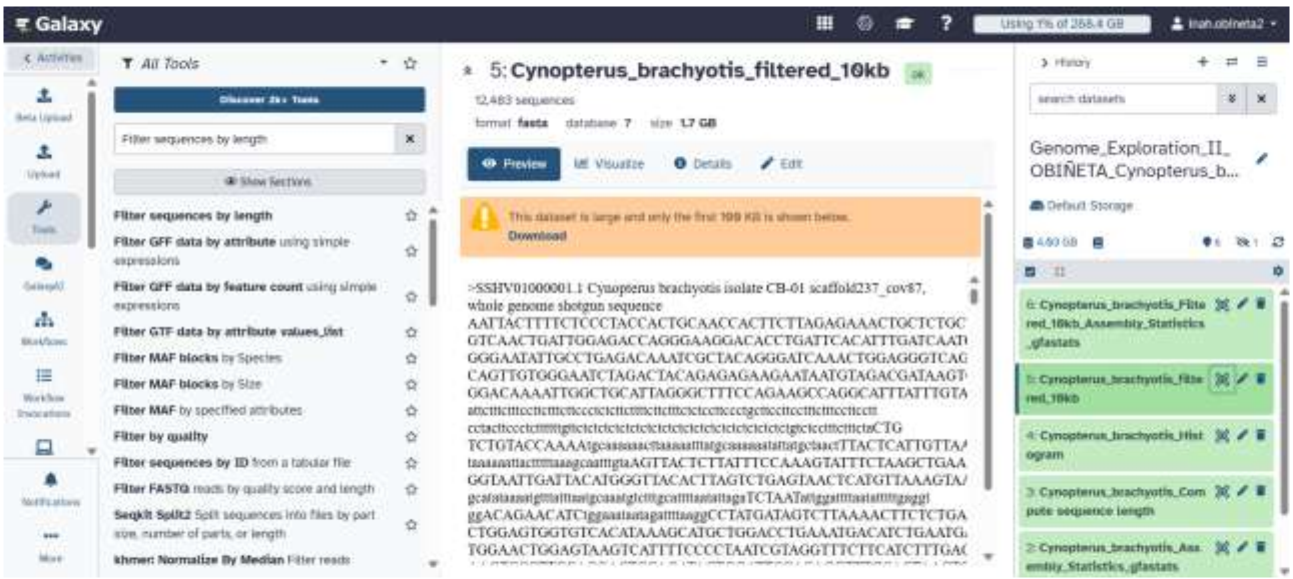

Figure 5: *Cynopterus brachyotis* Filtered 10kb Sequences Output using Filter Sequences by Length in Galaxy History

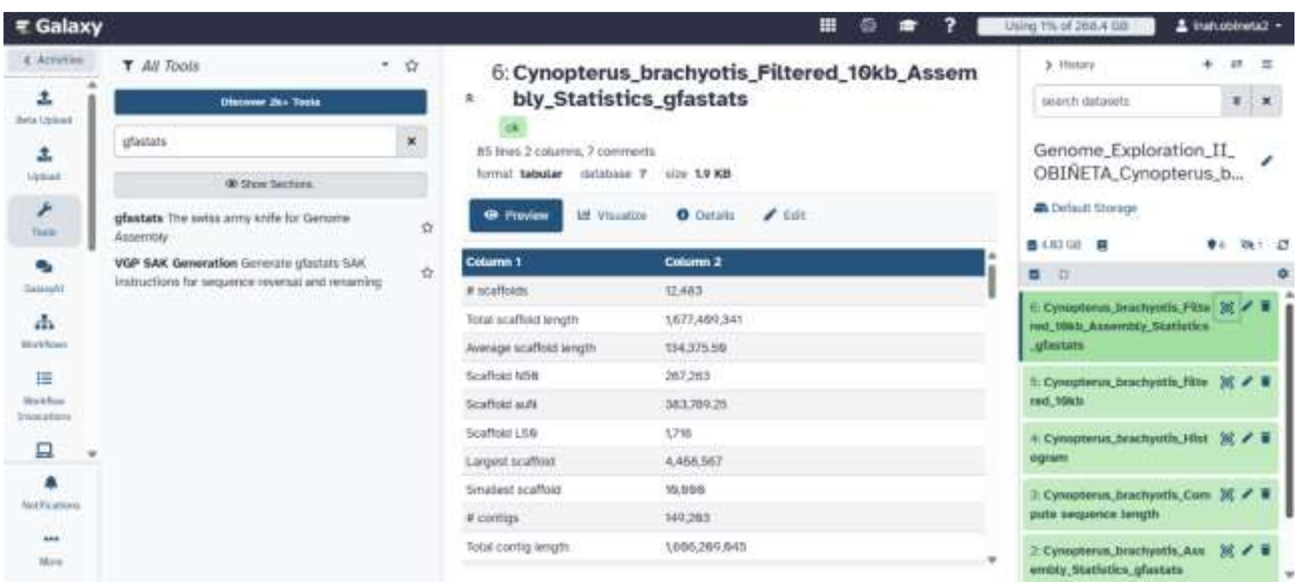

Figure 6: *Cynopterus brachyotis* Filtered 10kb Assembly Statistics Output rerun using gfastats in Galaxy History

- How many sequences were removed by the 10 kb filter?
    - 35,523 sequences were removed (48,006 − 12,483), leaving 12,483 sequences ≥10 kb
- Did the total assembly length decrease greatly or only slightly?
    - Only slightly where the total length dropped from 1,758,935,687 bp to 1,677,409,341 bp, a decrease of about 81.5 million bp (~4.6%), even though nearly 74% of the sequences were removed.
- Did N50 change? Why do you think it changed or stayed similar?
    - N50 rose slightly, from 251,278 bp to 267,263 bp. It removes all the short sequences and shifts the "midpoint" toward the longer ones that remain, so the metric looks a bit better even though no new data was actually added.
- What does this tell you about the contribution of short sequences to your assembly?
    - Most of the removed sequences were short that nearly three-quarters of the total count, but less than 5% of the genome's actual length. So, the real bulk of the genome lives in the longer scaffolds, while the short ones mostly add clutter, not substance. Still, they might carry real biological information, so they shouldn't be dismissed outright.

Part 5 — Small ORF Exploration

Metric	Value
Selected sequence	SSHV01000001.1
Sequence length	112,931 bp
Minimum ORF size used	300 bp
Number of ORFs found	40
Longest ORF	1,425 bp (SSHV01000001.1_21)

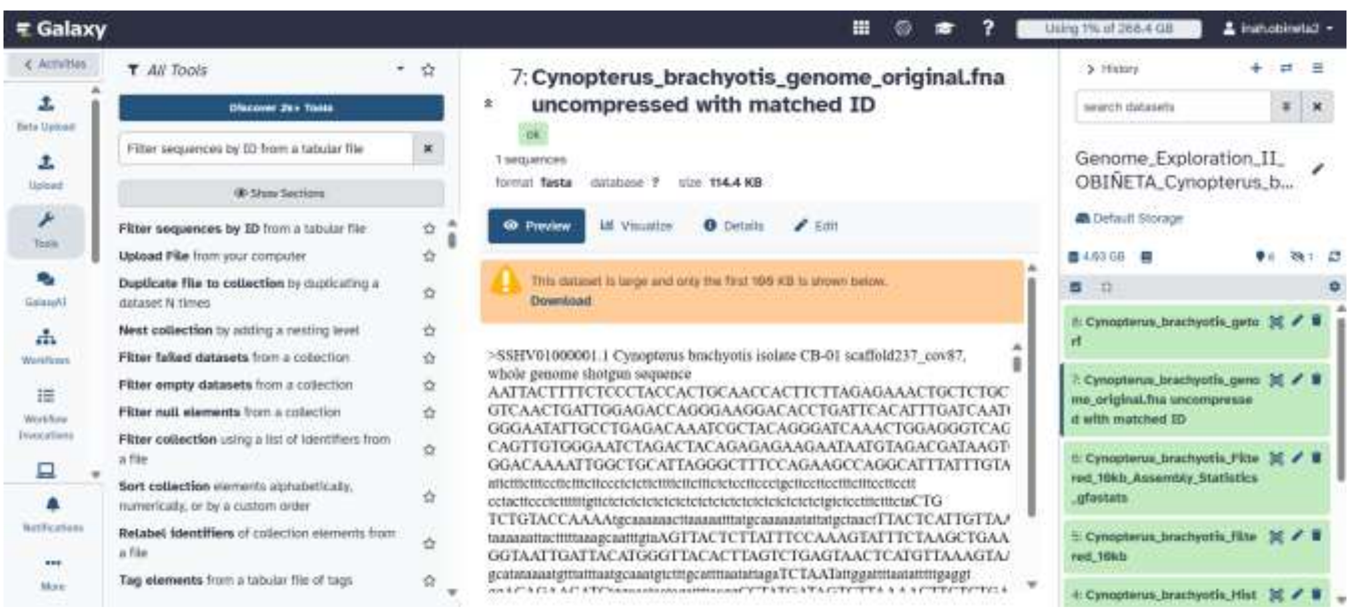

Figure 7: *Cynopterus brachyotis* Genome Original Uncompressed with Matched ID Output using the tool Filter sequence by ID in Galaxy History

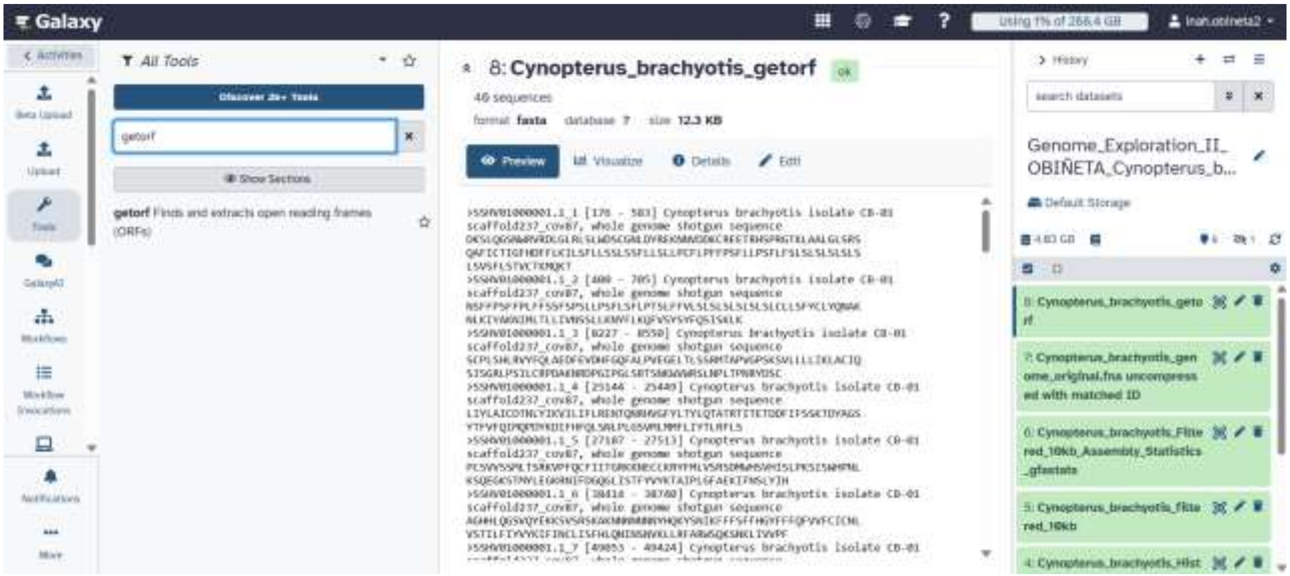

Figure 8: *Cynopterus brachyotis* ORF Prediction Output using getorf in Galaxy History

- How many ORFs were found in the selected sequence/region?
  - 40 ORFs were found in the selected 113 kb sequence (SSHV01000001.1)
- What was the longest ORF?
  - The longest ORF was 1,425 bp (SSHV01000001.1_21).
- Why should you NOT report every ORF as a real gene?
  - An ORF is only a sequence stretch without an internal stop codon. This means that the codon doesn't prove the region is actually transcribed, translated, or functional. Confirming a real gene requires further evidence like annotation, transcript data, or protein similarity.

Part 6 — Short Biological Interpretation

The *Cynopterus brachyotis* genome assembly is clearly fragmented rather than fully resolved into chromosomes. A high sequence count (48,006) relative to a modest maximum length (4.5 Mb) suggests the assembly process wasn't able to bridge most of the genome into large, continuous pieces which is likely a limitation of the original sequencing data rather than anything unusual about the bat's biology. The N50 (267,263 bp) and L50 (1,873) reinforce this where it takes nearly 2,000 scaffolds just to reach the halfway point of the genome, meaning no small set of sequences carries its bulk. Short sequences under 10 kb made up about 74% of the total count yet contributed less than 5% of the overall genome length shows that sequence count alone can be misleading, since most of the real biological content is concentrated in a smaller number of longer scaffolds. The GC content, at roughly 39%, falls within the normal range expected for a mammal, so nothing about the base composition stood out as unusual. Finally, testing ORF prediction on a 113 kb sequence returned 40 open reading frames, the longest at 1,425 bp demonstrates that stretches resembling coding regions are common, but an ORF alone isn't proof of a real gene. Structural prediction tools like getorf only flag possibilities; confirming actual function would require further evidence such as annotation or transcript data.

Part 7 — Galaxy Reproducibility and GitHub Documentation

GALAXY

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https://usegalaxy.org/u/inah.obineta2/h/genome-exploration-ii-obiñeta-cynopterus-brachyotis

GITHUB

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https://github.com/iinahmariee/OBINETA_assignment_02_genome_exploration

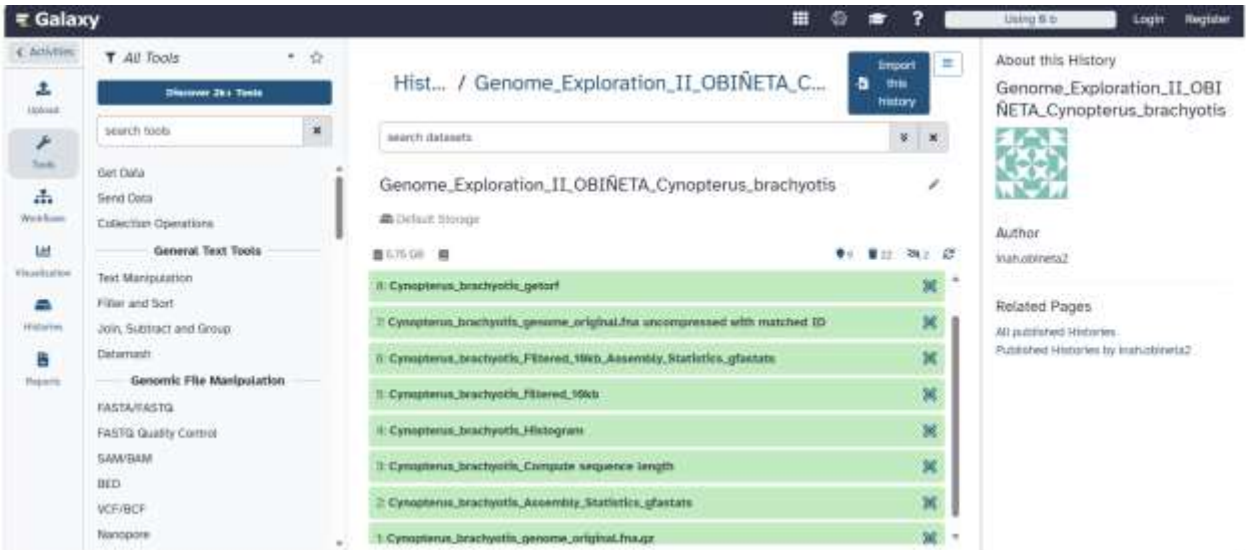

Figure 9: Workflow History of *Cynopterus brachyotis* Genome Exploration in Galaxy

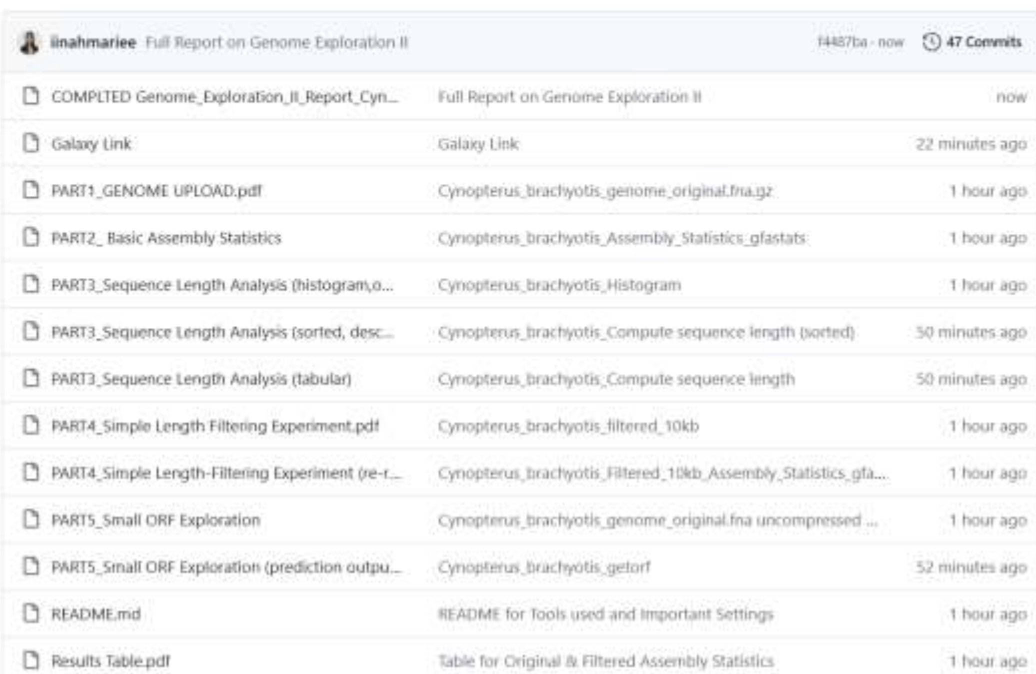

Figure 10: GitHub Repository for the activity